Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “kinetic energy” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

kinetic energy: energy associated with motion

final: at the end of the interval; subscript f

01 Warm-up: bring back an earlier idea

A force parallel to motion does **15 J** over **3 m**.
1. **Find its size**.
2. What work does the same force do with zero displacement?

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $W_{net} = \frac{1}{2}m(v_f^2 - v_i^2)$.
Constant mass. **Use the displacement** along each force component; stopping removes the initial kinetic energy.

Worked reasoning: A **2 kg** cart starts from rest. Constant net force **4 N** acts through **3 m**, doing **12 J**. Then $\frac{1}{2}mv^2 = \mathbf{12}$ gives $v = \sqrt{\mathbf{12}} \approx \mathbf{3.46\ m/s}$.

Your turn: 1. **Sketch a force diagram** and an energy bar diagram for this motion.
2. **Name where time** is unnecessary.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

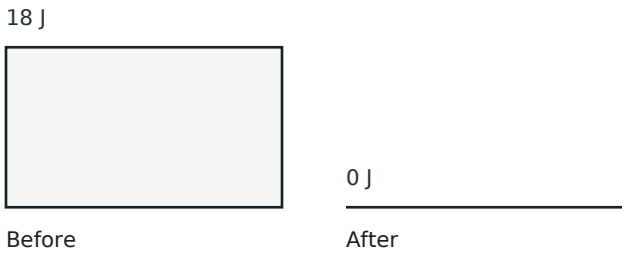
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cart has **18 J** kinetic energy and then stops. **How much** net work is done? **Show an energy bar before** and after.

Sentence starter: I notice ... . This supports ... because ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** lab cart moving at **6 m/s** meets a constant **3 N** resistive force. **Complete the energy account** and **find its stopping distance**. **Predict the new stopping distance** if its initial speed doubles; verify using kinetic energy.

Sentence starter: I predict … . I used … to make this prediction because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Initial momentum is **0.4 kg·m/s** and measured final momentum is **0.38 kg·m/s**. **Find percent deviation**. **Explain why this** is not itself a measured uncertainty. Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Before **After**

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A **5 kg** cart starts at rest. A **12 N** pull and **2 N** resistance act over **4 m**.
1. **Find final speed**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **4 kg** cart at **5 m/s** slows to **1 m/s** over **6 m**.
1. **Find the constant net force** component along its motion.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3